BIOCHEMISTRY CUMULATIVE EXAM

December 6, 1997

1. Many protein structures consist of two well-defined secondary structures. These are alpha-helices and beta-strands. List four independent features of an alpha-helix (its hydrogen-binding pattern, pitch per turn, handedness etc...) (10pt).

2. List four independent features of beta-sheets (there are three forms of the sheet structures known) (20pts).

3. Most peptide bonds are observed to be *trans*. If the residue that follows the peptide bond is ___, however, *cis* conformation is often observed. Fill in the blank (an amino acid residue) and draw the *cis/trans* isomers of the peptide unit consisting of the residue. For this peptide unit, describe why the *cis* isomer is energetically comparable to that of the *trans* isomer. (30pts)

4. Antibodies have been selected for their affinity for an antigen. An antibody is typically not an enzyme, but mankind has engineered them so that certain antibodies became catalytic. Why should a certain antibody ("catalytic-antibody") catalyze a given chemical reaction? Describe the principle (include the following: how would you make such an antibody; what antigen should you have in hand? etc....) (40pts).